

1. Given a string s which represents an expression, evaluate this expression and return its value.
Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as `eval()`.

Example 1:

Input: $s = "3+2*2"$

Output: 7

Example 2:

Input: $s = "3/2"$

Output: 1

Example 3:

Input: $s = "3+5 / 2"$

Output: 5

2. Given an array of integers. The task is to sort the given array on the basis of increasing count of distinct prime factors.

Input 1: $arr[] = \{30, 2, 1024, 210, 3, 6\}$

Output 1: 2 1024 3 6 30 210

Input 2: $arr[] = \{12, 16, 27, 6\}$

Output 2: 16 27 6 12

3. Given Two Arrays—1,2, 3,4, 5 and 2,3, 1,0, 5—Can You Find Which Number Is Not Present in the Second Array?

Example 1:

Input : $a[] = \{1, 2, 3, 4, 5, 10\}$; $b[] = \{2, 3, 1, 0, 5\}$;

Output : 4 10

Example 2:

Input : $a[] = \{92, 11, 493, 20, 5\}$; $b[] = \{5\}$;

Output : 92 11 493 20

4. Oneday there is a secret message that come from the Indian spy in USA. The message says, "There is a chemical bomb in USA which is capable of destroying the world completely". Indian spy also send the formula of that chemical bomb. Indian government call the most intelligent scientist of India Dr. sathish Kumar and give the assignment to find the anti-chemical solution only for the simple equations. But unfortunately he have very less time and not capable of finding the formula pattern of anti-chemical solution of complex equations. Help him to find the formula pattern of complex reaction by displaying the chemical reaction in the pattern format that is given below

Input format

The first line contain T the number of testcases

It is followed by T lines where each line contains a string ' S ' denoting the name of chemical reaction.

NOTE: The given string ' S ' contains only ODD LENGTH

Output

Print the pattern of given string for each testcase in a newline...

Sample Input:

1

ACIDIFICATION

Sample Output :

I I
C F F C
A I I A
T D D T
I I I I
O C C O
N A N
O C C O
I I I I
T D D T
A I I A
C F F C
I I

5. Write a program to rearrange the number in an array to generate a greatest possible number.

Input 1: [4,8,7,6,9,0]

Output 1: 987640

Input 2: [4,18,78,6,9,65,0]

Output 2: 9786654180